

Vertical-Slice Reconstruction in the Canonical Agler Space

A complete RKHS answer to Question 5.6 of arXiv:1312.0177

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Status: literature-implied answer (complete answer to Question 5.6). The needed mechanism is the classical restriction theorem for reproducing kernel Hilbert spaces. The new step here is its direct identification with the vertical-slice uniqueness argument already proved in the source paper.

1 Source question

On page 36 of [1], Bickel and Kneese ask:

Question 5.6. Assume $g \in \mathcal{H}(K_1^{\max})$. Is there an explicit way to construct g using only the function $g(0, w_2)$?

They add that a clean answer would give formulas for two canonical colligation blocks A and B .

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Question 5.6. Assume $g \in \mathcal{H}(K_1^{\max})$. Is there an explicit way to construct g using only the function $g(0, w_2)$?

A clean answer would also provide nice formulas for the operators A and B . It seems possible that the refined results in [13] about unitary T.F.R.'s associated to minimal augmented Agler decompositions might suggest methods of answering this question.

6. APPENDIX: VECTOR VALUED RKHS'S

In this section, we record several facts about vector valued reproducing kernel Hilbert spaces that were used in earlier sections. The

Figure 1: Question 5.6 and its stated motivation in the source PDF, page 36.

2 Identification and proof

The restriction principle used below is classical; see Aronszajn's Section 5, *The restriction of a reproducing kernel* [2]. We include the short operator-valued proof, so the conclusion does not depend on any unstated scalar-to-vector extension.

Let E_* be the coefficient Hilbert space, let

$$K = K_1^{\max} : \mathbb{D}^2 \times \mathbb{D}^2 \longrightarrow \mathcal{L}(E_*), \quad L(\zeta, \eta) := K((0, \zeta), (0, \eta)),$$

and let $R : \mathcal{H}(K) \rightarrow \mathcal{H}(L)$ be vertical restriction,

$$(Rg)(\zeta) = g(0, \zeta).$$

Here $\mathcal{H}(L)$ means the RKHS associated with the restricted kernel L , not an ambient Hardy norm.

Theorem 1 (Exact reconstruction from the vertical trace). *For the canonical kernel $K_1^{\max}$ of the source paper, R is a unitary operator. Hence, if $h(\zeta) = g(0, \zeta)$, then*

$$g = R^*h.$$

Equivalently, for every $z \in \mathbb{D}^2$ and $\nu \in E_*$,

$$\langle g(z), \nu \rangle_{E_*} = \langle h, \zeta \longmapsto K((0, \zeta), z)\nu \rangle_{\mathcal{H}(L)}. \quad (1)$$

Thus g is constructed using only its one-variable trace and the known canonical kernel.

Proof. For $\eta \in \mathbb{D}$ and $u \in E_*$, write

$$K_{(0, \eta)}u = K(\cdot, (0, \eta))u, \quad L_\eta u = L(\cdot, \eta)u.$$

On the dense span of the $L_\eta u$, define

$$J(L_\eta u) = K_{(0, \eta)}u.$$

The two Gram matrices agree:

$$\langle L_\eta u, L_\zeta v \rangle_{\mathcal{H}(L)} = \langle K_{(0, \eta)}u, K_{(0, \zeta)}v \rangle_{\mathcal{H}(K)}.$$

Therefore J extends isometrically from $\mathcal{H}(L)$ into $\mathcal{H}(K)$. Its range is

$$\overline{\text{span}}\{K_{(0, \eta)}u : \eta \in \mathbb{D}, u \in E_*\} = \{f \in \mathcal{H}(K) : f(0, \eta) = 0 \ \forall \eta \in \mathbb{D}\}^\perp.$$

The source paper proves immediately before Question 5.6 that the vertical slice is a uniqueness set for $\mathcal{H}(K_1^{\max})$. The vanishing space in the last display is consequently zero, so J is onto.

The reproducing property gives $R = J^*$, initially on kernel functions and hence everywhere. Since J is unitary, so are R and $R^* = J$, and $g = R^*(Rg)$. Finally,

$$\langle g(z), \nu \rangle = \langle g, K_z \nu \rangle_{\mathcal{H}(K)} = \langle h, RK_z \nu \rangle_{\mathcal{H}(L)},$$

while $(RK_z \nu)(\zeta) = K((0, \zeta), z)\nu$. This is (1). $\square$

3 Finite Gram reconstruction

Formula (1) is exact, but it also gives a concrete convergent algorithm. Choose points $\lambda_n \in \mathbb{D}$ with an accumulation point in $\mathbb{D}$, and choose a countable total set of vectors in E_* . Enumerate the resulting kernel sections as

$$\ell_j(\zeta) = L(\zeta, \lambda_j)u_j, \quad j \geq 1,$$

repeating points as needed to pair them with coefficient vectors. Let P_N denote the orthogonal projection in $\mathcal{H}(L)$ onto $\text{span}\{\ell_1, \dots, \ell_N\}$.

The coefficients $c_j^{(N)}$ of $P_N h = \sum_{j=1}^N c_j^{(N)} \ell_j$ are obtained from the finite Gram equations

$$\left\langle \sum_{j=1}^N c_j^{(N)} \ell_j, \ell_i \right\rangle_{\mathcal{H}(L)} = \langle h, \ell_i \rangle_{\mathcal{H}(L)} = \langle h(\lambda_i), u_i \rangle_{E_*}, \quad 1 \leq i \leq N.$$

If the Gram matrix is singular, its Moore–Penrose solution gives the same projection. Set

$$g_N(z) = \sum_{j=1}^N c_j^{(N)} K(z, (0, \lambda_j)) u_j. \quad (2)$$

The identity theorem and totality of the coefficient vectors imply that the ℓ_j 's have dense span in $\mathcal{H}(L)$. Consequently,

$$g_N = R^* P_N h \longrightarrow R^* h = g \quad \text{in } \mathcal{H}(K).$$

Bounded point evaluations give pointwise convergence, and local boundedness of $K(z, z)$ gives uniform convergence on compact subsets of $\mathbb{D}^2$. Every datum on the right side of the finite Gram equations is a value of the one-variable trace h .

4 The promised colligation-block formulas

Let $T = R^* : \mathcal{H}(L) \rightarrow \mathcal{H}(K)$ be the reconstruction operator in Theorem 1. In the notation of the source paper, for $f = (f_1, f_2)$ define the analytic one-variable functions

$$q_f(\zeta) = \begin{cases} \frac{f_1(0, \zeta) - f_1(0) + f_2(0, \zeta) - f_2(0)}{\zeta}, & \zeta \neq 0, \\ \partial_2 f_1(0) + \partial_2 f_2(0), & \zeta = 0, \end{cases}$$

and, for $\eta \in E$,

$$q_\eta(\zeta) = \begin{cases} \frac{\Phi(0, \zeta) - \Phi(0)}{\zeta} \eta, & \zeta \neq 0, \\ \partial_2 \Phi(0) \eta, & \zeta = 0. \end{cases}$$

The source paper proves that these traces belong to $\mathcal{H}(L)$. Its previously implicit unique functions are therefore

$$(Af)_2 = Tq_f, \quad (B\eta)_2 = Tq_\eta. \quad (3)$$

Their first components are then the source paper's structured Gleason quotients

$$(Af)_1(z) = \frac{f_1(z) - f_1(0) + f_2(z) - f_2(0) - z_2(Tq_f)(z)}{z_1},$$

$$(B\eta)_1(z) = \frac{(\Phi(z) - \Phi(0))\eta - z_2(Tq_\eta)(z)}{z_1},$$

with the removable values at $z_1 = 0$ filled in analytically. Combining (3) with (1) or the Gram scheme (2) gives the requested formulas using only vertical traces.

5 Scope and novelty status

This answers Question 5.6 for every coefficient Hilbert space covered by the source paper. It is not a new general RKHS theorem: the restriction mechanism is classical. The contribution of this note is the direct identification of the paper’s vertical uniqueness lemma with the injective case of that theorem, plus the explicit cross-kernel and finite-Gram reconstruction formulas.

A bounded search covered the run indexes; arXiv:1312.0177; the exact phrase of Question 5.6; the terms *canonical Agler*, *vertical slice*, *restriction*, and *reconstruction*; and later arXiv papers on Agler spaces and transfer-function realizations. No later paper explicitly answering Question 5.6 was found. Because the decisive abstract theorem is classical, the conservative provenance label is *literature-implied answer*, not a new full theorem. The official AMS metadata identifies Aronszajn’s Section 5 as the restriction theorem. An environment download limit prevented copying that supporting PDF into this packet; the source PDF and its question screenshot are included.

6 Review recommendation

Verify the convention for operator-valued kernel inner products in (1), the source paper’s vertical uniqueness argument immediately before Question 5.6, and that the trace data q_f, q_η are the ones in equations (5.8)–(5.9). The reconstruction theorem itself is a two-line adjoint argument once those points are fixed.

References

- [1] K. Bickel and G. Knese, *Canonical Agler Decompositions and Transfer Function Realizations*, Trans. Amer. Math. Soc. 368 (2016), 6293–6324; arXiv:1312.0177.
- [2] N. Aronszajn, *Theory of reproducing kernels*, Trans. Amer. Math. Soc. 68 (1950), 337–404, Section 5, doi:10.1090/S0002-9947-1950-0051437-7.